



Understanding behavioral intention to use a cloud computing classroom: A multiple model comparison approach



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ABSTRACT

Cloud computing is an innovative information technology that has been applied to education and has facilitated the development of cloud computing classrooms; however, student behavioral intention (BI) toward cloud computing remains unclear. Most researchers have evaluated, integrated, or compared only few theories to examine user BI. In this study, we tested, compared, and unified six well-known theories, namely service quality (SQ), self-efficacy (SE), the motivational model (MM), the technology acceptance model (TAM), the theory of reasoned action or theory of planned behavior (TRA/TPB), and innovation diffusion theory (IDT), in the context of cloud computing classrooms. This empirical study was conducted using an online survey. The data collected from the samples ($n = 478$) were analyzed using structural equation modeling. We independently analyzed each theory, by formulating a united model. The analysis yielded three valuable findings. First, all six theoretical models and the united model exhibited adequate explanatory power. Second, variance explanation, Chi-squared statistics, effect size, and predictive relevance results revealed the ranking importance of the theoretical models. Third, the united model provided a comprehensive understanding of the factors that significantly affect the college students' BI toward a cloud computing classroom. The discussions and implications of this study are critical for researchers and practitioners.

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1. Introduction

Innovation is one of the most critical forces in creating new services and products, developing new markets, promoting organizations' competitiveness, and transforming industries [30]. Cloud computing is an innovative technology that evolved from distributed, grid, and utility computing. Relevant products, such as mobile device applications including Gmail, Facebook, Twitter, YouTube, and Google Apps for Work, are proliferating [4] as more people use cloud computing services. Thus, cloud computing is a popular topic and global trend. This innovative technology comprises three types of services, namely infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS), providing diverse applications for customers [4,64]. IaaS encompasses the complete infrastructure required for

cloud computing, including virtual computers, servers, and storage devices (e.g., the Amazon S3 storage service and EC2 computing platform, and the Joyent, Terremark, and Rackspace cloud servers). PaaS provides computing models that run remotely on a platform, requiring hardware, an operating system, database, middleware, web servers, and other software (e.g., Salesforce's force.com, Microsoft's Azure services platform, Google App Engine, Amazon Relational Database Services, and Rackspace cloud sites). SaaS provides applications that run through the cloud; thus, users need not install any software (e.g., Salesforce, Google Apps for Work, and personal applications such as Gmail, Facebook, and Twitter) [4]. These three types of cloud computing services offer potential advantages including reduced costs, expected switching benefits, omnipresent services, collaborative support, access to infinite computing resources on demand, simplified operation, and increased use because of resource virtualization [4,52]. Seeking these advantages, many universities have implemented classroom-based cloud computing, called cloud computing classrooms, to enable students to learn from anywhere and at anytime [33,40,65]. Thus, a cloud computing classroom is defined as a ubiquitous learning environment that supports IaaS, PaaS, and

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SaaS in forms such as programs, objects, and websites and that can provide learning opportunities for individuals in and out of the physical classroom.

Theorists have attempted to explain and predict individual behaviors and have determined that behavioral intention (BI) is the dominant factor in the use of information systems (ISs) [72]. For example, the theory of reasoned action/theory of planned behavior (TRA/TPB) and technology acceptance model (TAM) are appropriate theories for explaining students' BI. In order to attract students to use cloud-based resources, student motivations should be considered. Motivational model (MM) theory can be used to assess student motivations. Cloud computing provides students with access to software and product services; therefore, students must be able to use these resources, and thus self-efficacy (SE) plays a critical role in their behavior. Service quality (SQ) and cloud services are also critical factors in the use of cloud computing classrooms. Thus, SE and SQ are suitable theories for explaining student behaviors. Cloud computing is an innovative technology that can be used to construct online classrooms and facilitate student learning. Innovation diffusion theory (IDT) is appropriate for investigating students' BI in the context of a cloud computing classroom. According to the preceding discussion, we focused on BI in the six theoretical models, namely the TRA/TPB, the TAM, the MM, SE, SQ, and IDT. Those who show a strong BI usually exhibit a correspondingly high level of use. Consequently, numerous studies have attempted to explain and predict BI [10,14,56,63]. However, these studies have typically applied only one to three theories to explain BI [14,63,75]. This method is limited to a complex phenomenon. Similarly, in the 19th century, the poet John Godfrey Saxe [61] wrote the poem *The Blind Men and the Elephant*, in which six blind men attempt to describe an elephant that they can feel, but not see. They conclude that the elephant is like a wall, spear, snake, tree, fan, or rope, depending on where they touch and engage in a heated debate that fails to yield the truth. Only by aggregating their descriptions can a comprehensive picture of the elephant be formed. In the context of cloud computing research, the elephant is BI and the blind people are the researchers who have attempted to empirically determine and explain BI by using a limited approach.

Furthermore, few studies have aggregated more than five theories to explain BI. For instance, Venkatesh et al. [72] developed a unified view of user intentions to use an IS and the consequent usage behavior, called the unified theory of acceptance and use of technology (UTAUT). Venkatesh et al. [72] reviewed and integrated constructs from the following eight theories and models: TRA, TAM, MM, TPB, a combined TPB and TAM (C-TPB-TAM), the model of PC utilization (MPCU), IDT, and social cognitive theory (SCT). In the cloud computing classroom context, cloud computing service is a focal point, and cloud computing efficacy is a critical factor in the initial learning stage of the cloud computing classroom. We provide an alternative view of user intention in contrast to UTAUT, particularly in cloud computing service by SQ theory and cloud computing efficacy by SE theory. Furthermore, Venkatesh et al. [72] used only variance (R^2) to compare the theoretical models. In our study, we used four criteria to evaluate the theoretical models: R^2 , Chi-squared (X^2) statistics, effect size (f^2), and predictive relevance (q^2). This study was aimed at developing an integrated view of intention to use cloud computing by reviewing and integrating numerous well-known theories, namely TRA/TPB, TAM, MM, SE, SQ, and IDT. This paper not only examines the effects of individual theories and the unified model on college students' intentions to use a cloud computing classroom, but also uses a multiple model comparison approach to empirically verify and examine their intentions. The following research questions are addressed: (a) Which theories or models most effectively elucidate BI in a cloud computing classroom? (b) What are the critical factors

of a unified model determining BI toward classroom-based cloud computing? The evaluated theories are compared and unified to elucidate BI. The remainder of this paper is structured as follows: Section 2 introduces the literature review; Section 3 details the research model and hypotheses; Section 4 presents the research methodology; Section 5 presents the data analysis and results; Section 6 provides a discussion, implications, and limitations; and Section 7 offers a conclusion.

2. Literature review

2.1. Cloud computing in the classroom

Educational organizations always seek opportunities to rationalize their resource management. Cloud computing is likely an immensely adoptable technology for many organizations because of its dynamic scalability and use of virtualized resources. For example, the University of Westminster in the United Kingdom has embraced Google Apps for Education, which provides free email, messaging, and shared calendars, and displays no advertisements. The Google platform also provides word processing, spreadsheet, and presentation support, facilitating collaboration on group assignments. Several other institutions of higher education in the United Kingdom (e.g., Leeds Beckett University, the University of Glamorgan, and the University of Aberdeen) have adopted Google Apps because of their low cost. In the United States, the University of California, Berkeley adopted Amazon web services to move its courses from the local infrastructure to the cloud. The Washington State University (Electrical Engineering and Computer Science) adopted the vSphere 4 cloud platform (VMware) to expand the services it offers to faculty and students. The vSphere 4 platform involves virtualization technology and is used to aggregate and manage IT resources, providing a seamless, flexible, and dynamic service with nearly limitless scalability. Cloud computing benefits educational institutions and has a significant impact in the classroom. For example, Stantchev et al. [65] investigated the motivations that lead higher education students to switch from using several learning management system (LMS) services for information sharing and collaboration to using cloud services. LMSs, also known as virtual learning environments, are like classrooms wherein they offer high levels of functionality regarding learning activities and features for course management and tracking. Cloud services encompass the functions of LMSs, enabling files to be stored and shared over the Internet through file synchronization. Stantchev et al. [65] reported that cloud hosting services were perceived as more user friendly than LMS services and that cloud services presented higher levels of perceived usefulness (PU) than the standard learning management tools. Lin et al. [40] studied a cloud-based learning environment aimed at developing students' self-reflection abilities to enable them to improve their learning motivation, comprehension, and performance. Conventional self-reflection methods are usually applicable only in classroom environments; however, cloud computing classrooms could be adopted for distance learning or after-class activities. Lin et al. determined that the cloud computing learning environment can effectively facilitate student reflection abilities and enhance their learning motivation, comprehension, and performance. Stein et al. [66] conducted a case study in rural high schools in North Carolina using the state's Virtual Computing Lab cloud service to access dynamic geometry and algebra software. The researchers found that a cloud service designed specifically for education can be applied to and improve K–12 education. Jou and Wang [33] studied how learning attitudes (ATTs) and academic performances were affected by the utilization of cloud computing technology, specifically computer-aided design (CAD) software. Students with a vocational high school

Table 1
Theoretical model comparisons.

Literature	Theories	Participants	Findings
Davis et al. [23]	TRA and TAM	107 students	The variance in intention explained by TRA was 32% and TAM was 47%.
Mathieson [45]	TAM and TPB	262 students	The variance in intention explained by TAM was 70% and TPB was 62%.
Taylor and Todd [68]	TAM and TPB (DTPB)	786 students	The variance in intention explained by TAM was 52%, and DTPB was 60%.
Plouffe et al. [53]	TAM and IDT	176 merchants	The variance in intention explained by TAM was 33% and IDT was 45%.
Chau and Hu [13]	TAM, TPB, and DTPB	408 professionals	The variance in intention explained by TAM was 40%, TPB was 32%, and DTPB was 42%.
Chau and Hu [14]	TAM and TPB	408 professionals	The variance in intention explained by TAM was 42%, TPB was 37%, and integrated model was 43%.
Premkumar and Bhattacharjee [56]	TAM and EDT	175 students	The variance in intention explained by TAM was 69%, EDT was 50%, and integrated model was 73%.
Shiau and Chau [63]	TAM and ECT	361 blog users	The variance in intention explained by TAM was 11%, ECT-IS was 46%, and integrated model was 47%.
Sun et al. [67]	TAM, TPB, and PMT	204 customers	The variance in intention explained by TAM was 32.6%, and TPB was 32.77%, and PMT was 38.8%.

background appeared to possess higher learning motivation for CAD applications than those who attended nonvocational high schools.

2.2. BI and model comparisons

Numerous studies have used well-known theories to predict and explain BI by comparing two or more models. For example, Chau and Hu [14] investigated the acceptance of telemedicine technology among physicians, comparing the TAM, the TPB, and an integrated model. Regarding the variance in intention, the results indicated that the TAM, TPB, and integrated model explained 42%, 37%, and 43% of the variance, respectively. Luo et al. [42] compared the MM and uses and gratifications (U&G) theory to evaluate web-based IS adoption. They used a partial least squares (PLS) analysis to test each theoretical model in an empirical setting, demonstrating that the MM, U&G theory, and integrated model explained 17.3%, 36.7%, and 43% of the variance in behavioral use, respectively. The UTAUT is another theory widely used for explaining BI and technology acceptance. Venkatesh et al. [72] developed the UTAUT to compare eight prominent theories, extending previous concepts to form a new research model that addressed facilitating conditions, performance expectancy, effort expectancy, social influence, BI, and user behavior. The moderating variables included gender, age, experience, voluntariness, and use. Venkatesh et al. compared eight prominent theories to predict the intention to use technology in a voluntary setting. The models explained the following amount of variance in intention: TRA = 30%, TAM/TAM2 = 38%, MM = 37%, TPB/DTPB = 37%, combined TAM and TPB (C-TAM-TPB) = 39%, MPCU = 37%, IDT = 38%, SCT = 37%, and UTAUT = 40% (Ref. [72], pp. 440, 462). Table 1 summarizes the previous theoretical model comparisons.

In sum, cloud computing brings real benefits for educational institutions and university students. Cloud computing classrooms provide powerful functions and flexibility to university students. Because of these advantages, more universities are adopting cloud computing classrooms, and the administrators of these universities are seeking to understand students' behavior. A theoretical model may be used to explain a certain behavior. Previous studies have applied only one to three theories to explain BI [14,63,67]. This study not only tests individual theoretical models but also proposes a unified model for explaining students' BI to use a cloud computing classroom.

3. Research model and hypotheses

A cloud computing system was established at a university in Northern Taiwan with more than 18,000 students. During the initial stage of establishing cloud computing classrooms, the

university moved some computer laboratory functions to the university cloud. The cloud classroom provides SaaS, PaaS, and IaaS services, including cloud folders, cloud hosting, educational software, Techficiency Quotient Certification training files, Microsoft Office, Adobe Creative Suite, programming tools, and specialized applications such as statistical software and tools. The managers of this university have attempted to realize students' motivations, SE, acceptance, and planned behavior toward cloud computing classrooms. The MM, SE, TAM, and TRA/TPB are appropriate for explaining the phenomena related to motivations, SE, acceptance, and planned behavior. Moreover, previous studies have confirmed that six well-known theories, namely TRA/TPB, TAM, MM, SE, SQ, and IDT, have strong predictive and explanatory power regarding user intention [72,74].

A good theory should explain phenomena with few constructs, such as a parsimonious model. All the six well-known theories effectively explain the phenomena of a cloud computing classroom. For example, a TRA with personal ATTs and subjective norms toward intention, which was proposed by Fishbein and Ajzen [24], has become more prominent, receiving considerable attention in the human behavior field [62]. The TPB is an extension of the TRA that adds a construct of perceived behavioral control [2]. Perceived behavioral control is theorized to be an additional determinant of intention and behavior. The TPB has been used to elucidate individual intentions and behaviors toward diverse technologies [45,68]. Thus, a TPB with ATTs, subjective norms, and perceived behavioral control are considered in the context of a cloud computing classroom. Davis et al. [23] used the TRA to investigate the individual acceptance of technology and proposed the TAM. This model, comprising the two core constructs of PU and perceived ease of use (PEOU), has been widely applied in the IS literature to study individual intentions and behaviors in the contexts of various information technology (IT) such as personal computers, computer applications, [20,45] the Internet, blogs [63], and advanced mobile phone services [31]. Motivation theory with intrinsic and extrinsic motivation has been used to determine the crucial factors driving human activities and to predict and explain human intention and behaviors [70]. For example, Davis et al. [22] applied motivational theory to understand the adoption and use of new technology. The authors associated PU (an extrinsic motivation) with performance as a consequence of use according to the reinforcement and enjoyment of the process (an intrinsic motivation) of performing a behavior. Motivation theory with playfulness (intrinsic motivation) and PU (extrinsic motivation) is considered in the current study in the context of a cloud computing classroom. SE refers to perceived personal confidence when undertaking particular tasks or challenges in specific contexts [6]. SE can be assessed at domain- or task-specific levels, and such measures may demonstrate strong validity and predictive

relevance [51]. In the context of a cloud computing classroom, computer SE (CSE) and cloud SE (OSE) are used to measure personal judgments of the ability to use computers [18,44] or cloud computing applications in diverse situations. SQ involves comparing expectations with performance to measure how well a delivered service conforms to client expectations. In the context of a cloud computing classroom, most applications are in the SaaS category. Assessments of application SQ (ASQ) and overall cloud service are used to measure the individual perceptions of the SQ of a cloud computing classroom. Since IDT was introduced in marketing in the 1960s, numerous studies have used the theory as a theoretical framework to examine the intention to use IT and IT adoption and use [72]. The relative advantage construct in IDT is often considered as the equivalent of the PU construct in the TAM, and the complexity construct in IDT is also similar to the PEOU construct in the TAM [15]. Observability of IDT is used for measuring visible innovations; in this study, it is replaced with visibility (VIS). Result demonstrability and voluntariness are considered as perceptions affecting the adoption of an IT innovation [47]. According to IDT and in the context of a cloud computing classroom, compatibility, result demonstration (RD), trialability, VIS, and voluntariness are considered as innovative factors that determine the students' BI toward a cloud computing classroom. According to the preceding review and discussion of the literature, we propose a united model of BI by incorporating six well-known theories: the TRA/TPB, the TAM, the MM, SE, SQ, and IDT (Fig. 1).

Perceived behavioral control was added to the TRA [24] to develop the TPB [1]. Both the TPB and TRA are used to elucidate human behaviors by identifying and analyzing the determinants of BI [1,24]. Studies have validated and supported the relationships among the TRA and TPB constructs. For example, ATTs and subjective norms (i.e., the TRA) significantly influence intention [2]. Perceived behavioral control is a critical factor determining user intention [2,38]. Intention typically predicts and explains behavior [2,38]. In the context of the cloud computing classroom, the users believe the degree of control to perform a behavior (perceived behavior control, PBC). Users may perceive pressure from others to study or not study in a cloud computing classroom, which constitutes a subjective norm. The ATT determines the positive or negative assessments of the users regarding executing a behavior in a cloud computing classroom. According to the TPB and considering the cloud computing classroom context, we hypothesize the following:

H1. *Perceived behavioral control is positively associated with the intention to study in a cloud computing classroom.*

H2. *Subjective norms are positively associated with the intention to study in a cloud computing classroom.*

H3. *Attitude is positively associated with the intention to study in a cloud computing classroom.*

Applying the TRA, Davis [21] proposed the TAM for studying computer acceptance behaviors. The TAM yields strong predictions and explanations for diverse ISs, including computer applications, enterprise resource planning, digital libraries, and e-shopping systems [20,45,63,67,69]. The TAM posits that PU and PEOU are the critical determinants of system use. PU represents the extent to which a person believes that using a specific application system improves his or her job performance. PEOU is the degree to which a person believes that using a particular system would be effortless [23]. PU and PEOU are distinct dimensions linked to ATTs and use. Various studies have shown that the ATT is a weak predictor of the intention to use [23,68]. Some studies have excluded ATT-based constructs; instead a parsimonious and simple TAM comprising the constructs PEOU, PU, and BI has been used [56,71,72]. PU directly affects user intention [23,31,56,63,71,72]. PEOU is positively associated with PU [63,71] and BI [63,67,71,72]. This study focused on user intention. Thus, we hypothesize the following:

H4a. *PEOU is positively associated with the intention to study in a cloud computing classroom.*

H4b. *PEOU is positively associated with the PU of studying in a cloud computing classroom.*

H4c. *PEOU is positively associated with the attitude toward studying in a cloud computing classroom.*

H5a. *PU is positively associated with the intention to study in a cloud computing classroom.*

H5b. *PU is positively associated with the attitude toward studying in a cloud computing classroom.*

The MM involves the use of intrinsic or extrinsic motivations to explain human behaviors. Numerous researchers have posited that BI can be both extrinsically and intrinsically motivated. For example, Venkatesh et al. [73] redefined the TAM within a motivational framework, suggesting that both extrinsic and intrinsic motivations predict BI to use technology. From an extrinsic motivational perspective, BI is driven by perceived values and benefits. PU explains the utility value of using a system and is a key driver of BI to use (e.g., H5a: PU is positively associated with the intention to study in a cloud computing classroom). From an intrinsic motivational perspective, behaviors are performed to derive feelings of fun, happiness, and pleasure. Perceived enjoyment (perceived playfulness) occurs in the current context when people perceive that using a computer is enjoyable; this is a form of intrinsic motivation. Numerous studies have demonstrated that perceived enjoyment critically influences user intention. For example, Moon and Kim [46] investigated Internet use, and determined that PU and perceived playfulness significantly and positively affect BI. Furthermore, Lee et al. [36] studied the acceptance of Internet-based learning mediums, and found that PU and perceived enjoyment significantly and positively affect BI. Regarding our study on the intention to use cloud-computing classrooms, if users perceive that using cloud computing is

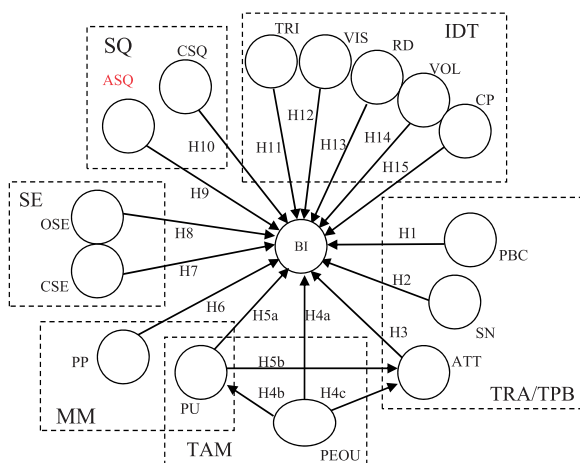


Fig. 1. United model of behavioral intention. ATT: attitude; CP: compatibility; CSE: computer self-efficacy; BI: behavioral intention; OSE: cloud self-efficacy; CSQ: cloud service quality; PBC: perceived behavior control; PEOU: perceived ease of use; PP: perceived playfulness; PU: perceived usefulness; RD: result demonstration; SN: subjective norm; ASQ: application service quality; TRI: trialability; VIS: visibility; VOL: voluntariness.

enjoyable, then they are likely to demonstrate an increased intention to use this technology. Thus, we hypothesize the following:

H6. *Enjoyment is positively associated with the intention to use a cloud computing classroom.*

According to the SCT [7], SE reflects a person's belief in his or her ability to attain particular levels of performance. Multon et al. [48] meta-analyzed various SE studies, which revealed significant relationships between SE and performance and also showed that SE can predict performance. Because of the rapid development of IT, SE was extended to CSE, which reflects a person's judgment of his or her capability to use a computer [18]. Numerous CSE studies have demonstrated that as employee performance increases, computer-induced anxiety decreases, leading to promotions [18,43]. CSE has been divided into two dimensions: general CSE, which is used to assess the general beliefs of users regarding their computer skills (e.g., confidence in using software to complete a task), and SE for a specific application, which is used to assess the level of confidence in using specific applications (e.g., confidence in using Microsoft Excel, PowerPoint, or Word) [43,44]. In the context of cloud computing classrooms, CSE is used to assess the student's confidence level in using software skills to complete a task. In a cloud computing classroom, specific application SE (i.e., cloud self-efficacy, OSE) refers to the ability to use cloud-based applications. In general, people who consider themselves competent computer users are likely to use computers [50]. The higher SE people exhibit, the more likely they are to complete tasks. Similarly, people who exhibit high general CSE or OSE believe that they can perform well and are likely to intend to use cloud computing classrooms. Thus, we hypothesize the following:

H7. *High general CSE positively affects the intention to use a cloud computing classroom.*

H8. *High OSE positively affects the intention to use a cloud computing classroom.*

Customers form service expectations according to their past experiences, word of mouth, and advertisements; SQ is used to assess and compare perceived and expected services. SQ is traditionally applied to offline environments that facilitate personal contact. Numerous studies have used SQ to predict and assess customer reactions and responses, such as willingness to pay a premium price and purchase additional products or services, and to determine customer satisfaction levels [19,59]. Because of the advancement of IT, assessing SQ is critical in the relatively new domain of online business, in which firms deliver products and services through web channels. Because IT provides the medium for delivering the service [26], SQ is assessed according to customers' overall evaluations of services and applications provided through a website. The importance of SQ has been stressed in the IS field because of the increasing number and type of services delivered using websites [12,74]. In the context of a cloud computing classroom, SQ is assessed according to the overall evaluations of students regarding cloud SQ (CSQ) and ASQ. In addition, previous research has demonstrated associations between SQ and specific dimensions of BI [3,34,35]. Thus, we hypothesize the following:

H9. *High ASQ positively affects the intention to use a cloud computing classroom.*

H10. *High CSQ positively affects the intention to use a cloud computing classroom.*

In practice, innovation and diffusion are critical characteristics of products and services that have gained substantial academic attention. The diffusion of innovation has been widely examined in

disciplines such as marketing, education, sociology, communication, agriculture, and IT. Rogers [60] defined IDT as the process by which an innovation is communicated through channels over time among the members of a social system. Within the framework of Rogers [60], IDT involves five characteristics of innovation: relative advantage, compatibility, complexity, trialability, and observability. Liang and Lu [39] investigated the factors influencing the willingness of the public to adopt online tax filing services, classifying current users into early and late adopters. The results showed that trialability and observability significantly influenced the adoption intentions of late adopters but not those of early adopters. Relative advantage, compatibility, and complexity significantly influenced the adoption intentions of current users. Lee et al. [37] combined IDT and the TAM to study the factors affecting the BI of business employees toward using e-learning systems. The results indicated that compatibility, complexity, relative advantage, and trialability significantly affected PU. Furthermore, complexity, relative advantage, and trialability significantly affected PEOU. All five perceptions of innovation characteristics (relative advantage, compatibility, complexity, trialability, and observability) significantly influenced the BI of employees to use e-learning systems. In addition, Venkatesh et al. [72] regarded trialability, VIS, result demonstrability, voluntariness, and compatibility as important factors affecting user intention. Cloud computing is a new technology, and cloud computing classrooms are innovative learning system environments; these IDT innovation characteristics are suitable for evaluating the BI of students toward using a cloud-computing classroom. Thus, we hypothesize the following:

H11. *Trialability positively affects BI to use a cloud computing classroom.*

H12. *Visibility positively affects BI to use a cloud computing classroom.*

H13. *Result demonstrability positively affects BI to use a cloud computing classroom.*

H14. *Voluntariness positively affects BI to use a cloud computing classroom.*

H15. *Compatibility positively affects BI to use a cloud computing classroom.*

4. Research methodology

Structural equation modeling (SEM) is a crucial multivariate data analysis method adopted in many fields including marketing research, education, IS, and organizational science. Researchers use SEM to assess latent variables at the observational level and test the relationships between the latent variables at the theoretical level. SEM comprises covariance-based SEM (CB-SEM) and variance-based PLS-SEM. Although these techniques involve distinct approaches, they share the same roots [32]. CB-SEM is used to minimize the discrepancies between the estimated and sample covariance matrices according to the estimated model parameters; this model requires making multivariate normality assumptions. PLS-SEM is used to estimate the partial model relationships in an iterative sequence of ordinary least squares regressions, maximizing the explained variance of the endogenous latent variables and relaxing the multivariate normality assumptions. Numerous studies, including those conducted by Chin and Newsted [17], Gefen et al. [27], and Hair et al. [29], have compared the approaches. CB-SEM has traditionally been used to estimate models and is a useful form of theoretical testing in diverse

disciplines [27,29]. PLS-SEM has been recognized as an alternative to CB-SEM in research for prioritizing prediction. PLS-SEM is used because it involves a complex model setup, explains the variance of endogenous constructs, and enables exploratory and theory development, the use of nonnormal data and small sample sizes, and the formative measurement of latent variables [17,29]. The weaknesses of CB-SEM are the strengths of PLS-SEM. When applying SEM techniques, researchers must consider the research objectives, data characteristics, and model types [27,57]. PLS-SEM is suitable for analyzing the complex model in this study, which unites six theories and 16 constructs. When there is a lack of appropriate measures for the overall model fit, PLS-SEM is limited in comparing alternative model structures [29]. CB-SEM was suitable for evaluating each individual theoretical model and comparing the six theoretical models. Both CB-SEM and PLS-SEM will be used to support our research objective, clarifying user intention by comparing and unifying six well-known theories. SPSS version 19.0 was used to measure the descriptive statistics. SmartPLS version 2.0 M3 (PLS-SEM) was used to estimate an overall model unifying the six theories [58]. LISREL version 8.8 (CB-SEM) was used to estimate each theoretical model and compare and rank the six theories.

4.1. Participants

The research models were tested using data collected from the users of a cloud computing classroom. In order to compare and unify the six theoretical models, a field study was conducted, evaluating a medium-size university, which established the first cloud computing classrooms in Taiwan. A two-part online survey was used to test the proposed theoretical models. The first part comprised questions measuring 16 constructs about the research models and the second part captured demographic data regarding the participants, who were assured that their personal information would remain confidential. Of the 488 returned web survey questionnaires, 10 exhibited incomplete data, yielding 478 valid responses for use in the data analysis.

Because some students did not complete the survey, nonresponse bias might be a concern. Armstrong and Overton [5] suggested that late respondents, compared with early respondents, are more likely to resemble nonrespondents. Comparison of the gender and the age of the early and late respondents using the *t* test indicated no significant differences ($p > 0.05$). Thus, we excluded the possibility of nonresponse bias. In addition, because all the data were collected from a single source at the same time, common method variance might be a concern [55]. We used a two-step procedure to minimize common method bias, specifically by guaranteeing respondent anonymity and refining questionnaire items through pretesting [8,55]. Furthermore, we assessed the dataset using Harman's one-factor test to identify any potential common method bias [54]. No general factor accounted for more than 50% of the variance, suggesting that the common method bias was not a concern.

4.2. Measurement development

In this study, we focused on six theories (TRA/TPB, the TAM, the MM, SE, SQ, and IDT), and 16 constructs were adapted from previous studies. Each construct was operationalized as a reflective model. According to Fishbein and Ajzen, the ATT represents how willing or unwilling a person is to use a cloud computing classroom [24]. The authors suggest that subjective norms are operationalized as a person's perception that most of the people who are valuable to him or her think that he or she should or should not use the cloud computing classroom [24]. BI refers to the subjective probability that a person will use the

cloud computing classroom [24]. Perceived behavioral control refers to the perceived ease or difficulty of using a cloud computing classroom [2]. PU is defined as the subjective perception of a user that using the cloud computing classroom will yield enhanced academic achievement [23]. PEOU refers to the degree to which the user expects using the cloud computing classroom to be effortless [23]. Compatibility is defined as the degree to which an innovation is perceived as being consistent with the existing values, needs, and past experiences of users regarding cloud computing classrooms. Voluntariness refers to the degree to which using the cloud computing classroom is perceived as voluntary. Result demonstrability is defined as the tangibility of the results of using the cloud computing classroom. VIS refers to the degree to which a person observes others using the cloud computing classroom. Trialability refers to the degree to which users can try or practice using the cloud computing classroom [47]. Perceived playfulness is the strength of the belief that interacting with the cloud computing classroom will fulfill various intrinsic motives [46]. CSQ refers to an overall service evaluation of the cloud computing classroom [11,76]. ASQ refers to the degree to which the key functionalities of the software used in the cloud computing classroom meet the requirements of college students [9]. OSE refers to personal self-confidence in the ability to use a cloud computing classroom [68]. CSE refers to the personal judgment regarding the ability to use multiple computer applications [43]. The measurement items were adapted from related studies and slightly modified to suit the context of a cloud computing classroom. The scale items were scored on a five-point Likert scale that ranged from 1 (*strongly disagree*) to 5 (*strongly agree*). The primary survey was conducted after determining the content validity of the questionnaire. Appendix A contains a summary of the measurement items.

5. Results

5.1. Demographic profiles

Descriptive statistics indicated that 51% of the participants were male ($N = 244$), 49% female ($N = 234$), 68% 18–22 years old, 30.4% 21–23 years old, and 1% 24–26 years old. The participants reported the following amount of experience using cloud computing classrooms: 1–2 months (45%), 2–5 months (21.8%), 5–6 months (6.3%), 6–7 months (3.7%), or >7 months (23.2%). The amount of time spent using cloud computing classrooms was <30 min (38.3%), 30 min to 1 h (33.9%) 1–2 h (18.4%), 2–3 h (6.1%), or >3 h (3.3%).

5.2. Measurement model

A measurement model was used to assess the reliability and validity of the study. Fornell and Larcker [25] suggested evaluating measurement scales as follows: (a) all indicator factor loadings should be significant and exceed 0.5, (b) construct reliabilities should exceed 0.8, and (c) the average variance extracted (AVE) by each construct should exceed the amount of measurement error variance ($AVE > 0.5$). The results indicated that all indicator loadings exceeded 0.5 (range: 0.61–0.94), all construct reliabilities exceeded 0.8 (range: 0.88–0.95), and all AVEs exceeded 0.50 (range: 0.7–0.86), indicating satisfactory convergent validity. The discriminant validity was determined by calculating the square root of the AVE for each construct exceeding the correlation between the other constructs [16]. The results listed in Table 2 show that all criteria were met, indicating that the proposed models demonstrate satisfactory reliability and validity.

Table 2
Discriminate validity of research model.

	ATT	CP	CSE	BI	OSE	CSQ	PBC	PEOU	PP	PU	RD	SN	ASQ	TRI	VIS	VOL
ATT	0.93															
CP	0.50	0.91														
CSE	0.37	0.39	0.90													
BI	0.58	0.60	0.30	0.87												
OSE	0.59	0.55	0.47	0.60	0.88											
CSQ	0.47	0.58	0.42	0.54	0.51	0.93										
PBC	0.47	0.54	0.42	0.60	0.57	0.40	0.89									
PEOU	0.60	0.62	0.43	0.64	0.63	0.53	0.69	0.93								
PP	0.41	0.59	0.41	0.50	0.49	0.69	0.41	0.47	0.91							
PU	0.60	0.66	0.40	0.67	0.62	0.55	0.59	0.75	0.50	0.88						
RD	0.50	0.63	0.41	0.64	0.66	0.53	0.60	0.62	0.56	0.65	0.91					
SN	0.53	0.59	0.36	0.54	0.43	0.47	0.48	0.49	0.50	0.50	0.45	0.84				
ASQ	0.52	0.57	0.55	0.52	0.59	0.62	0.53	0.55	0.60	0.59	0.61	0.47	0.87			
TRI	0.47	0.62	0.45	0.55	0.59	0.64	0.51	0.56	0.67	0.60	0.58	0.53	0.66	0.87		
VIS	0.32	0.51	0.37	0.51	0.49	0.44	0.49	0.48	0.56	0.44	0.56	0.53	0.50	0.55	0.91	
VOL	0.52	0.59	0.42	0.57	0.65	0.48	0.56	0.63	0.47	0.63	0.64	0.46	0.55	0.56	0.48	0.84

ATT: attitude; CP: compatibility; CSE: computer self-efficacy; BI: behavioral intention; OSE: cloud self-efficacy; CSQ: cloud service quality; PBC: perceived behavior control; PEOU: perceived ease of use; PP: perceived playfulness; PU: perceived usefulness; RD: result demonstration; SN: subjective norm; ASQ: application service quality; TRI: trialability; VIS: visibility; VOL: voluntariness.

5.3. Structural model

Each of the six individual theories explained the following percentages of variance in BI: TPB = 62% ($R^2 = 0.62$; degree of freedom [df] = 59), TAM = 66% ($R^2 = 0.66$; df = 24), MM = 69% ($R^2 = 0.69$; df = 24), SE = 48% ($R^2 = 0.48$; df = 32), SQ = 48% ($R^2 = 0.48$; df = 32), and IDT = 66% ($R^2 = 0.66$; df = 155). Table 3 contains the beta path coefficient and R^2 value of each theory.

The unified model of the six theories (Fig. 2) explained 61.8% ($R^2 = 0.618$) of variance in BI.

Table 3
Beta and R^2 of each theory.

Models	Independent variables	Dependent variables: BI	
		Beta	R^2
TPB	Perceived behavior control (PBC)	0.39***	0.62
	Subjective norm (SN)	0.17***	
	Attitude (ATT)	0.36***	
TAM	Perceived ease of use (PEOU)	0.20**	0.66
	Perceived usefulness (PU)	0.63***	
	Perceived playfulness (PP)	0.68***	
MM	Perceived ease of use (PEOU)	0.20**	0.69
	Perceived usefulness (PU)	0.63***	
	Perceived playfulness (PP)	0.68***	
SE	Computer self-efficacy (CSE)	-0.03	0.48
	Cloud self-efficacy (OSE)	0.71***	
	Application service quality (ASQ)	0.38***	
SQ	Cloud service quality (CSQ)	0.38***	0.48
	Triability (TRA)	0.07	
	Visibility (VIS)	0.13**	
IDT	Relative advantage (RD)	0.34***	0.66
	VOL Voluntariness (VOL)	0.20**	
	Compatibility (CAB)	0.20***	

Note: **($p < 0.01$); ***($p < 0.001$).

Table 4
Theoretical effect sizes for f^2 and q^2 .

	Behavioral intention (BI)			
	f^2	Effect size	q^2	Effect size
IDT	0.0485	Small	0.0233	Small
TPB	0.0950	Small	0.0706	Small
TAM	0.1147	Small	0.0455	Small
MM	0.0444	Small	0.0231	Small
SE	0.0416	Small	0.0207	Small
SQ	0.0244	Small	0.0119	–

5.4. Theoretical effect size

In addition to evaluating the R^2 values of BI, the change in the R^2 value when a specified exogenous construct is omitted from the model was used to evaluate whether the omitted construct has a substantial impact on BI. This measure is referred to as the f^2 effect size. The effect size can be calculated as

$$f^2 = \frac{R^2_{\text{included}} - R^2_{\text{excluded}}}{1 - R^2_{\text{included}}}$$

In addition to evaluating the magnitude of the R^2 values as a criterion of predictive accuracy, we examined the Stone–Geisser Q^2 value [28]. In the structural model, Q^2 values > 0 for BI indicate the path model's predictive relevance for this particular construct. The relative impact of predictive relevance can be compared by means of the measured q^2 effect size as follows:

$$q^2 = \frac{Q^2_{\text{included}} - Q^2_{\text{excluded}}}{1 - Q^2_{\text{included}}}$$

We extended an exogenous construct to main constructs of a theory, where R^2_{included} and R^2_{excluded} are the R^2 values of the BI when

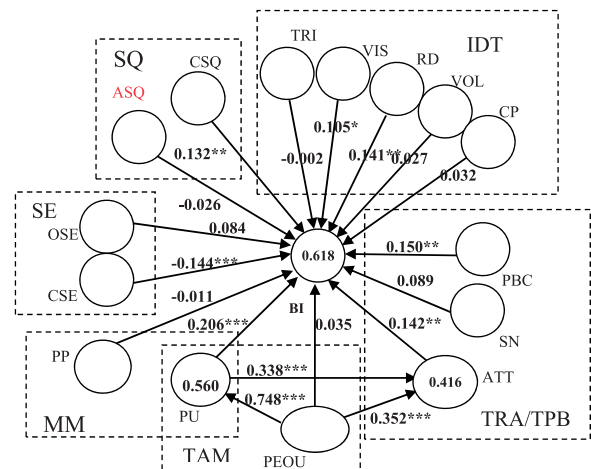


Fig. 2. Results of the unified model. ATT: attitude; CP: compatibility; CSE: computer self-efficacy; BI: behavioral intention; OSE: cloud self-efficacy; CSQ: cloud service quality; PBC: perceived behavior control; PEOU: perceived ease of use; PP: perceived playfulness; PU: perceived usefulness; RD: result demonstration; SN: subjective norm; ASQ: application service quality; TRI: trialability; VIS: visibility; VOL: voluntariness.

the selected main constructs of a theory are included in or excluded from the model. The change in the R^2 values is calculated by estimating the PLS path model twice. It is estimated first with the main constructs of a theory included (yielding $R^2_{included}$) and then with the main constructs of a theory excluded (yielding $R^2_{excluded}$). While assessing f^2 and q^2 , values of 0.02, 0.15, and 0.35 represent small, medium, and large effects [28]. Table 4 summarizes the theoretical effect size results for f^2 and q^2 .

6. Discussion

6.1. Discussion

This study combined six user intention-related theories to develop a unified model for explaining BI to use a cloud computing classroom. The analysis yielded three key findings.

First, all the six theoretical models (the MM, the TAM, IDT, the TPB, SE, and SQ) exhibited strong explanatory power of intention to use cloud computing classroom, with R^2 values ranging from 0.48 to 0.69, indicating that the theories are capable of providing an insight into the cloud computing classroom behavior. All the factors of the theoretical models, except CSE and trialability, exert significantly positive effects on the intention to use a cloud computing classroom. According to the TPB results, cloud computing classrooms are designed to facilitate study by college students. Initially, the university provides software applications and services through cloud computing classrooms. After cloud computing classroom promotion and training, college students use the software applications and services. If the students perceive cloud computing classrooms as useful, they will use them more frequently and other students will begin using them. According to the TAM results, students' use of cloud computing classrooms may increase his or her learning performance. The more advantages students perceive cloud computing classrooms as offering, the more likely they are to use cloud computing classrooms. The cloud computing classroom is an innovative means of facilitating student learning. Students must become skilled at using cloud computing classrooms. The less effort that using a cloud computing classroom requires, the more likely students are to use it. According to the MM results, the cloud computing classroom facilitates course content learning by college students. Professors can use the cloud computing classroom to assign novel and engaging tasks that will increase the willingness of students to use the system. According to the SE results, having skills and knowledge related to using cloud computing classroom software applications and services makes college students more likely to use the system. Our results indicate that CSE is not a significant factor for determining the intention to use cloud computing classroom, possibly because students learn basic computer skills in their first year of college. Because they already possess computer skills, they do not consider CSE an important factor in using cloud computing classrooms. According to the SQ results, both general SQ and ASQ are significant factors for determining the intention to use cloud computing classrooms. Cloud computing classrooms are accessible anywhere and anytime; any college student with an Internet connection can use the cloud computing classroom services when encountering a learning problem or find an answer to a course-related problem. According to the IDT results, the innovative characteristics of the cloud computing classrooms include compatibility, voluntariness, result demonstrability, VIS, and trialability. College students prefer cloud computing classroom applications that are compatible with those on their PCs and thus require less effort to use. Furthermore, students enjoy sharing homework and exercise results with classmates, which is a means of developing friendships. If using a cloud computing classroom benefits college students, they do not require an external force to push them to use the system. However,

college students dislike the work required to master various new applications. They may like to learn new skills from others such as professors, which requires less time and effort.

Second, different criteria yielded different results regarding BI toward a cloud computing classroom. According to the comparison of the R^2 results, the MM exhibited the greatest variance explanatory power ($R^2 = 0.69$), followed by the TAM ($R^2 = 0.66$), IDT ($R^2 = 0.66$), the TPB ($R^2 = 0.62$), and SE and SQ ($R^2 = 0.48$). The comparisons of F statistics with R^2 and df values yielded similar results, with the MM and TAM exhibiting the strongest explanatory power, followed by IDT, the TPB, and SE and SQ. The MM and TAM, which focus on motivation, had the greatest explanatory power; the innovative characteristics of IDT also exhibited strong explanatory power. The TPB focuses on self-control and also explains cloud computing classroom behavior well. Finally, SE and SQ, which focus on the ability and service aspects, had the least explanatory power. The comparison of theoretical effect size f^2 showed that the TAM had the greatest effect size, followed by the TPB, IDT, the MM, SE, and SQ. The comparisons of effect size q^2 yielded similar results, with the TPB exhibiting the strongest effect size, followed by the TAM, IDT, the MM, SE, and SQ. For all theories, the TPB and TAM have larger effect size compared to the other models. Consequently, IDT and MM have larger effect size than SE and SQ. In summary, using different analysis criteria yielded different results.

Third, a unified model effectively explains cloud computing classroom behavior ($R^2 = 0.618$) and provides more comprehensive viewpoints. According to the united model of the six theories, PU had the strongest positive effect on user intention, followed by ATT, CSQ, PBC, RD, VIS, and OSE; the effects of all these factors were significant. The factors PEOU, PP, ASQ, trialability, voluntariness, compatibility, and subjective norms did not exert significant effects on user intention. CSE had significantly negative effects on user intention, possibly because cloud computing classrooms move the functional software of PCs to the cloud. College students with strong computer skills can perform their school work on their personal computers; thus, acquiring higher CSE in order to shift to using cloud computing classroom software and services has not gained much attention.

6.2. Theoretical and practical implications

The current findings yield various theoretical and practical implications in the user behavior domain. Theoretically, these results confirm that each of the six theories (the MM, the TAM, IDT, the TPB, SE, and SQ) used to explore BI toward cloud computing classrooms demonstrated adequate explanatory power. Using one theory derives only one perspective of cloud computing classroom use. Integrating multiple theories sheds light on crucial phenomena and clarifies critical factors in a comprehensive model. Our results also confirm that the unified model has an adequate explanatory power to explain BI toward cloud computing classrooms. According to these results, researchers may focus on developing context-specific antecedents to the established constructs in this unified model of cloud computing classrooms.

In practice, enterprises may leverage the benefits of cloud computing classroom including omnipresent services, easy to use, collaborative support, and infinite computing resources on demand. Moreover, a cloud computing classroom is a ubiquitous learning environment that supports IaaS, PaaS, and SaaS in forms such as programs, objects, and websites and that can provide learning opportunities for internal training or staff development. For internal training, a cloud computing classroom may offer innovative learning and knowledge solutions for staffs including courseware, content, and toolkits to assist their work efficiently. Staffs may learn how to enhance their capability, support

customers more effectively, and build a solid strategy to enable their long-term growth with partnership through sales support, partner services, knowledge sharing, and advanced insights into the latest industry developments. Staffs may also maximize knowledge transfer by providing an unrivaled learning experience. For staff development, a cloud computing classroom may support staffs to develop their carrier, such as staff identifying issues that he or she would like to learn. A cloud computing classroom may also record how his or her work has been carried out. The omnipresent services of a cloud computing classroom can ensure recognition of an individual's contribution to the enterprise; these provide opportunities for staffs to discuss about the training needs for the current role and any development for future career, and also regarding the difficulties or obstacles that hamper effectiveness and the required solutions. In sum, understanding and leveraging the advantages of cloud computing classrooms may yield more insightful guidance for practitioners.

6.3. Limitations and future research

Although the six individual theoretical models and unified model adequately explained the BI to use cloud computing classrooms, our findings have four main limitations. First, the empirical data were collected at a university. Additional data, such as commercial or industrial data, may require further verification. Second, because the respondents were current cloud computing classroom users, generalizations about cloud computing classroom discontinuance are beyond the scope of this study. Thus, future research efforts may also consider the phenomenon of cloud computing classroom discontinuance. Third, each of the six theoretical models adequately explained cloud computing classroom. Future studies should investigate the antecedents and consequences of these models according to the characteristics of cloud computing classrooms. Fourth, unifying the six well-known theories may not sufficiently elucidate the cloud computing classroom, and future studies should consider incorporating additional theories associated with behavior in cloud-based classrooms. Researchers should also focus on parsimonious and comprehensive points of view depending upon the specific contexts and/or distinct research objectives in the future studies.

7. Conclusion

The advancement of the Internet and computational evolution has produced innovative IT and cloud computing services. Colleges and universities provide cloud computing systems as a novel service to attract students. Thus, understanding the BI of students toward cloud computing classrooms is vital. Data were collected from a medium-size university. Both covariance-based SEM (conducted using LISREL) and component-based SEM (performed using PLS analysis) were used to test the empirical data. The six individual theoretical models and unified model demonstrated strong explanatory power regarding the BI to use a cloud computing classroom. However, each theoretical model exhibited distinct features that could make it superior depending upon the context and research objective. The unified model provided a comprehensive view of the factors affecting the BI to use cloud computing classrooms. We clarified this BI by comparing and unifying six well-known theories (the TRA/TPB, the TAM, the MM, SE, SQ, and IDT) in the context of a cloud computing classroom. The analysis yields three findings. First, we offer four criteria for evaluating the theoretical model comparisons, namely R^2 , X^2 , f^2 , and q^2 . Comparison of the R^2 and X^2 values showed that the MM and TAM were the most effective theoretical models for elucidating BI. Moreover, a comparative study of f^2 and q^2 values revealed that the TAM and TPB had larger effect sizes than the other models. Second, we elucidate the critical factors affecting BI toward cloud computing classrooms. According to the unified model, the factors PU, ATT, CSQ, PBC, RD, VIS, and OSE exerted significantly positive effects on the college students' intention to use a cloud computing classroom. Third, our results may serve as a valuable reference to managers when planning, evaluating, and implementing systems to provide classroom-based cloud computing. All the six theoretical models and the unified model exhibited an adequate explanatory power of intention to use a cloud computing classroom. We accept the notion that every theoretical model has distinct features that make it superior to others in specific contexts and according to different research objectives. The unified model provides a comprehensive view of the factors affecting the intention to use a cloud computing classroom.

Appendix A

Construct	Measurement items	Adapted from
Perceived behavioral control	PBC1. I would be able to handle the cloud computing classroom. PBC2. Using the cloud computing classroom is entirely within my control. PBC3. I have resources, knowledge, and the ability to make use of the cloud computing classroom.	Taylor and Todd [68]
Subjective norms	SN1. People who influence my behavior would think that I should use the cloud computing classroom. SN2. People who are important to me would think that I should use the cloud computing classroom. SN3. My classmates would think that I should use the cloud computing classroom. SN4. My professors would think that I should use the cloud computing classroom.	Taylor and Todd [68]
Attitude	ATT1. Using the cloud computing classroom is a good idea. ATT2. Using the cloud computing classroom is a wise idea. ATT3. I like the idea of using the cloud computing classroom.	Taylor and Todd [68]
Perceived ease of use	PEOU1. Instructions for using applications in the cloud computing classroom will not be hard to follow. PEOU2. It will be difficult to learn how to use the cloud computing classroom. PEOU3. It will be easy to operate the applications in the cloud computing classroom.	Taylor and Todd [68]
Perceived usefulness	PU1. Using the cloud computing classroom will improve my grades. PU2. The advantages of the cloud computing classroom will outweigh the disadvantages. PU3. Overall, using the cloud computing classroom will be advantageous.	Taylor and Todd [68]
Perceived playfulness	PP1. When interacting with cloud computing classroom, I do not realize the time elapsed PP2. While interacting in a cloud computing classroom, I am not aware of any noise. PP3. Using cloud computing classroom gives enjoyment to me for my task.	Moon and Kim [46]
Computer self-efficacy	CSE1. I believe I have the ability to install new software applications on a computer. CSE2. I believe I have the ability to identify and correct common operational problems with a computer.	Marakas et al. [43]

Appendix A (Continued)

Construct	Measurement items	Adapted from
Cloud self-efficacy	CSE3. I believe I have the ability to unpack and set up a new computer. OSE1. I believe I have the ability to do tasks in a cloud computing classroom if there was no one around to tell me what to do. OSE2. I believe I have the ability to finish tasks in a cloud computing classroom if I had only the software manuals for reference. OSE 3. I believe I have the ability to finish tasks in cloud computing classroom if I had seen someone else using it before trying it myself. OSE4. I believe I have the ability to finish tasks in a cloud computing classroom if I could not call someone for help as I got stuck.	Compeau and Higgins [18]
Application service quality	ASQ1. Data reporting and extracting features in a cloud computing classroom are available. ASQ2. The configuration (e.g., user administration, etc.) features of an application are available in a cloud computing classroom. ASQ3. An application's help functionalities are provided in a cloud computing classroom. ASQ4. An application's core features support the process steps/activities in a cloud computing classroom.	Benlian et al. [9]
Cloud service quality	CSQ1. The quality of services of cloud computing classroom is excellent CSQ2. The quality of services of cloud computing classroom is superior CSQ3. The quality of services of cloud computing classroom is high standards	Oh [49]
Triability	TRI1. I have had a great deal of opportunity to try various cloud computing classroom applications. TRI2. I know where I can go to satisfactorily test various services of the cloud computing classroom. TRI3. The cloud computing classroom was available to me to adequately test run various applications. TRI4. Before deciding whether to use any cloud computing classroom applications, I was able to properly try them out.	Moore and Benbasat [47]
Visibility	VIS1. I have seen what others do using their cloud computing classroom applications. VIS2. In my school, one sees a cloud computing classroom on many desks. VIS3. I have seen a cloud computing classroom in use outside my school. VIS4. It is easy for me to observe others using a cloud computing classroom in my school.	Moore and Benbasat [47]
Result demonstrability	RD1. I would have no difficulty telling others about the results of using a cloud computing classroom RD2. I believe I could communicate to others the consequences of using the cloud computing classroom. RD3. The results of using the cloud computing classroom are apparent to me.	Moore and Benbasat [47]
Voluntariness	VOL1. My professors do not expect me to use a cloud computing classroom. VOL2. My use of a cloud computing classroom is voluntary. VOL3. My professors do not require me to use a cloud computing classroom.	Moore and Benbasat [47]
Compatibility	CP1. Using a cloud computing classroom is completely compatible with my current situation CP2. I think that using a cloud computing classroom fits well with the way I like to study. CP3. Using a cloud computing classroom fits into my study.	Moore and Benbasat [47]
Behavioral intention	BI1. I intend to use the cloud computing classroom to print projects, papers, or assignments this term. BI2. I intend to use the cloud computing classroom frequently this term. BI3. I will recommend cloud computing classroom to others.	Taylor and Todd [68] Lu et al. [41]

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